Stress Prediction Using Personalized Pre-Sleep Lifelog Features

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***Abstract*— This study proposes a pre-sleep stress prediction model combining raw behavioral and environmental features, personalized Z-score features, and recent questionnaire and sleep history using the DACON/ETRI lifelog dataset. In nested chronological validation, the proposed model improved ROC-AUC from 0.560 to 0.663 and reduced Log Loss from 0.654 to 0.610. Ablation analysis showed that history features provided most of the gain, while Z-score features offered complementary information. Nested LOSO validation also improved pooled ROC-AUC from 0.381 to 0.643 for held-out participants with at least five days of history.**

Keywords—lifelog data, stress prediction, personalized features, Z-score, sleep history

I. Introduction

Stress is closely associated with sleep quality, fatigue, and mental health, making continuous stress monitoring increasingly important for preventive healthcare and personalized digital health services [1].

Recent advances in smartphones and wearable devices enable continuous collection of multimodal lifelog data, including physiological, behavioral, and environmental information. The ETRI Lifelog Dataset 2024 provides a realistic benchmark for stress prediction by combining multimodal sensing data with subjective questionnaires on stress, fatigue, and sleep quality [9].

Previous studies have shown that wearable physiological signals and smartphone behavioral data are effective for stress recognition [3]-[6]. However, many existing approaches rely on laboratory settings, population-level models, or daily aggregated features, which may overlook individual behavioral baselines and recent changes in personal state. Deep learning models can provide competitive predictive performance, but their limited interpretability can restrict their use in personalized digital-health services [10].

To address these limitations, this study proposes an interpretable pre-sleep stress prediction framework using the ETRI Lifelog Dataset 2024. The framework combines raw behavioral and environmental features extracted from the 30-min period before bedtime, personalized Z-score features, and recent questionnaire and sleep-history features. Regularized logistic regression was trained using nested chronological validation, while participant-clustered bootstrap and nested leave-one-user-out validation were used to assess uncertainty and cross-user generalization.

The main contributions of this study are as follows

- An interpretable pre-sleep stress prediction framework combining raw, personalized Z-score, and recent questionnaire and sleep-history features is proposed.
- The contributions of Z-score and history features are examined through ablation analysis.
- Missing sensor observations are retained using training-only imputation and data-availability indicators.
- Performance is evaluated using nested chronological validation, participant-clustered bootstrap, and nested LOSO validation.

The following research questions are addressed.

RQ1. Do questionnaire and sleep-history features improve prediction beyond raw behavioral and environmental features?

RQ2. Do personalized Z-score features provide additional predictive information?

RQ3. Does the proposed model generalize to held-out participants with available personal history?

II. Related Work

A. Lifelog-based Mental Health Prediction

Recent advances in smartphones and wearable devices have enabled passive collection of multimodal lifelog data for continuous mental health monitoring. StudentLife demonstrated that smartphone sensing data are associated with perceived stress [3]. More recently, multimodal datasets integrating behavioral, environmental, and physiological information have enabled more realistic stress prediction in everyday environments [1], [9].

B. Personalized Stress Prediction

Because physiological and behavioral characteristics differ across individuals, personalized stress prediction has become an important research direction. Tazarv et al. [2] demonstrated that user-specific modeling improves prediction performance compared with population-level approaches. Previous studies have mainly focused on physiological baselines or user-specific models, while the complementary roles of behavioral deviations and recent questionnaire and sleep history have received less attention.

C. Research Gap and Motivation

Although previous studies have shown promising results, they often rely on laboratory settings, population-level models, or daily aggregated features, limiting their applicability to personalized stress prediction in real-world environments [2], [4]-[6]. Recent deep learning approaches using the same ETRI dataset achieve competitive performance but offer limited interpretability [10].

To overcome these limitations, this study proposes an interpretable stress prediction framework based on raw pre-sleep behavioral and environmental features, personalized Z-score representations, recent questionnaire and sleep history, and regularized logistic regression with nested validation.

TABLE I. COMPARISON WITH PREVIOUS STUDIES

Study	Dataset	Model	Main Strength
MIS-LSTM [10]	ETRI	CNN-LSTM	High Predictive Performance
Proposed	ETRI	Regularized Logistic Regression	Interpretable and Personalized

III. Feature Design and Proposed Model

A. Feature Design

Features represented absolute pre-sleep behavior, deviations from individual baselines, and recent questionnaire and sleep history. Behavioral and environmental variables were extracted from the 30-min period before bedtime, while physiological variables were excluded.

The candidate pool included 36 raw features related to screen use, ambient light, activity, movement, and device use, together with 34 personalized Z-score features:

$$z_{i,t,f} = \frac{x_{i,t,f} - \mu_{i,t-1,f}}{\sigma_{i,t-1,f} + \epsilon}$$

The mean and standard deviation were calculated using only observations before date t. Twenty history features were derived from previous Q1–Q3 and S1–S4 values, including previous values, three-day means, and prior means. All history variables were shifted by one day to prevent leakage. Three additional features represented sensor availability and missingness. Interaction terms were excluded because they did not improve performance.

B. Feature Selection and Prediction Model

Feature reduction was performed independently within each training fold. Features with more than 95% missing values or near-zero variance were removed, and highly correlated variables ($|r| \geq 0.95$) were treated as redundant. The remaining features were ranked using coefficients from L2-regularized logistic regression, and the top 20 or 30 features were selected through inner validation.

Logistic regression with L2 or Elastic Net regularization was used **[7].** Feature count, regularization strength, Elastic Net ratio, and class weighting were selected by inner-validation Log Loss. The main comparison was between the Raw baseline and the proposed Raw+Z+History model, while Raw+Z and Raw+History were evaluated through ablation analysis.

C. Probability Calibration

Platt scaling was applied as a post-hoc probability-calibration method using only inner out-of-fold predictions [12]. The calibrated probability was defined as $p_{cal} = \sigma(a\text{logit}(p) + b)$

The fitted calibrator was then applied to the outer-test predictions. Uncalibrated probabilities were used for the primary discrimination analysis, while calibrated probabilities were evaluated using Log Loss, Brier score, and reliability diagrams.

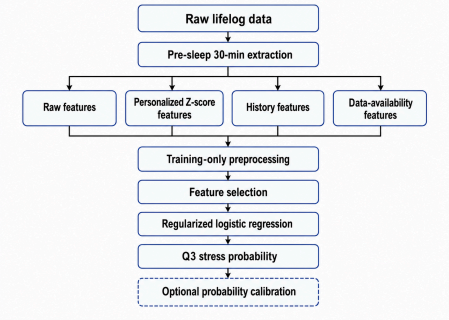


Fig. 1. Pipeline for Stress Prediction

IV. Data and Analysis Methods

A. Dataset and Preprocessing

The DACON/ETRI Lifelog Dataset 2024 [9], released through the 4th ETRI Human Understanding AI Paper Challenge on DACON [11], includes smartphone- and wearable-derived data, daily questionnaires, and sleep outcomes from 10 participants. The target was the binary pre-sleep stress indicator Q3.

The original metric table contained 450 participant-day observations, including 180 non-stress and 270 stress cases. Because the 30-min sensor-feature table contained only 244 matched observations, sensor features were left-joined to the metric table by participant ID and sleep date. The remaining 206 observations were retained with missing sensor values. Requiring at least five previous days of history resulted in 400 eligible observations, including 217 pooled outer-test predictions.

All preprocessing was estimated from training data only. Continuous variables were clipped to the training-set 1st and 99th percentiles, missing values were replaced with the training median, and missingness indicators were added. Features were then standardized using the training-set mean and standard deviation.

B. Nested Chronological Validation

The primary evaluation used nested chronological expanding-window validation. In each outer fold, earlier observations were used for training and the subsequent period for testing. Pooled performance across all 217 outer-test predictions was used as the primary estimate.

Within each outer training set, an additional chronological validation selected features, hyperparameters, class weighting, and probability calibration. The outer-test data were not used during model development.

ROC-AUC and PR-AUC assessed discrimination [8], while Log Loss and Brier score evaluated probability accuracy. Balanced Accuracy and Specificity were also reported.

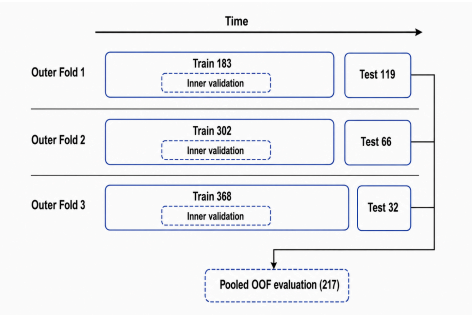


Fig. 2. Nested chronological validation

C. Sensitivity and Ablation Analyses

Class-weighted and unweighted models were compared using inner Log Loss. Minimum-history requirements of 3, 5, and 7 days were also examined, with 5 days used as the primary condition.

Ablation analysis compared Raw, Raw+Z, Raw+History, and Raw+Z+History under identical preprocessing and validation procedures. These comparisons were treated as exploratory secondary analyses without formal multiple-comparison correction.

D. Bootstrap and LOSO Validation

Performance uncertainty was estimated using 5,000 participant-clustered bootstrap resamples. Participants were sampled with replacement, and 95% confidence

intervals were obtained from the 2.5th and 97.5th percentiles.

Cross-user generalization was evaluated using nested leave-one-user-out validation. One participant was held out in each outer iteration, while feature and model settings were selected using an inner user-level leave-one-out procedure. The held-out participant's previous questionnaire and sleep records could be used to construct history features, but none of that participant's observations were used for model fitting. Thus, this setting represented transfer to a held-out participant with at least five days of personal history rather than a fully history-free cold start.

V. Results

A. Chronological Prediction Performance

TABLE II. CHRONOLOGICAL NESTED VALIDATION RESULTS

Model	ROC-AUC	PR-AUC	Bal. Acc.	Specificity	Log Loss
Raw	0.560	0.736	0.511	0.083	0.654
Raw+Z+History	0.663	0.787	0.615	0.486	0.610

The primary evaluation used 217 pooled out-of-fold observations from 10 participants. Compared with the Raw baseline, the proposed Raw+Z+History model improved ROC-AUC from 0.560 to 0.663 and PR-AUC from 0.736 to 0.787. Log Loss decreased from 0.654 to 0.610, and Specificity increased from 0.083 to 0.486.

Participant-clustered bootstrap analysis with 5,000 resamples confirmed significant improvements in ROC-AUC (+0.103, 95% CI:0.017–0.200), Log Loss (−0.045, 95% CI:−0.078 to −0.006). PR-AUC and Balanced Accuracy also improved numerically, but their confidence intervals included zero. Therefore, ROC-AUC and Log Loss were treated as the primary performance measures.

B. Ablation Analysis

An ablation analysis was conducted to separate the contributions of personalized Z-score and history features.

TABLE III. ABLATION ANALYSIS OF FEATURE GROUPS

Feature set	ROC-AUC	PR-AUC	Bal. Acc.	Specificity	Log Loss
Raw	0.560	0.736	0.511	0.083	0.654
Raw+Z	0.542	0.734	0.493	0.000	0.650
Raw+History	0.651	0.774	0.612	0.458	0.612
Raw+Z+History	0.663	0.787	0.615	0.486	0.610

Adding Z-score features alone did not improve ROC-AUC, whereas adding history features increased ROC-AUC from 0.560 to 0.651 and reduced Log Loss from 0.654 to 0.612. Adding Z-score features to Raw+History produced a further, smaller improvement. Thus, recent questionnaire and sleep history provided the main predictive gain, while personalized behavioral deviations contributed complementary information.

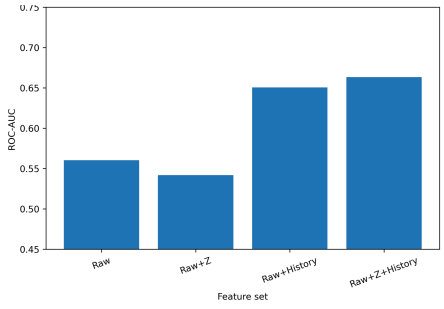


Fig. 3. Comparison of ROC-AUC across feature sets

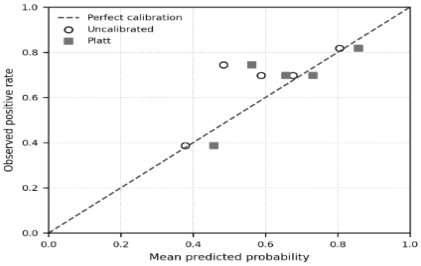
C. Probability Calibration and Fold-Wise Performance

Platt scaling was fitted using only inner out-of-fold predictions.

TABLE IV. PROBABILITY CALIBRATION OF THE PROPOSED MODEL

Calibration	*ROC-AUC*	*PR-AUC*	*Log Loss*	*Brier Score*
Uncalibrated	0.663	0.787	0.610	0.212
Platt scaling	0.669	0.791	0.593	0.203

Calibration reduced Log Loss and Brier score, and the reliability diagram showed probabilities closer to the ideal diagonal. Uncalibrated results were used for the main



discrimination analysis, while Platt-scaled results were reported as an additional assessment of probability reliability.

Fig. 4. Reliability diagram of the proposed model before and after Platt scaling using five common quantile bins

TABLE V. PERFORMANCE ACROSS CHRONOLOGICAL OUTER FOLDS

Fold	ROC-AUC	PR-AUC	Log Loss
1	0.641	0.732	0.654
2	0.693	0.825	0.575
3	0.734	0.862	0.517

Performance increased numerically across later folds. However, Fold 3 contained only 32 test observations, compared with 119 and 66 in the preceding folds. Therefore, the pooled 217 out-of-fold predictions were treated as the primary estimate.

D. Sensitivity Analysis

Balanced class weighting increased Specificity from 0.472 to 0.556, but decreased Balanced Accuracy from 0.615 to 0.598 and increased Log Loss from 0.613 to 0.630. Thus, class weighting improved non-stress detection at the cost of probability accuracy, and the unweighted setting was retained based on inner Log Loss.

TABLE VI. SENSITIVITY TO THE MINIMUM NUMBER OF HISTORICAL DAYS

Minimum History	Evaluation N	ROC-AUC	Log Loss
3 days	225	0.661	0.614
5 days	217	0.661	0.613
7 days	207	0.700	0.604

The 3-day and 5-day settings produced similar results, while the 7-day setting showed higher numerical performance. Because each condition used different observations, the 7-day result was interpreted as a sensitivity trend rather than evidence of an optimal threshold. The 5-day setting was retained for the primary analysis.

E. Stable Feature Analysis

Features with non-zero coefficients and consistent directions across all three chronological folds were considered stable.

TABLE VII. STABLE FEATURES ACROSS CHRONOLOGICAL FOLDS

Feature	Type	Folds	Median β	Direction
Previous Q3	History	3/3	0.317	+
Previous Q1	History	3/3	0.213	+
Light change	Raw	3/3	0.206	+
3-day sleep efficiency	History	3/3	−0.102	−
Still-activity ratio Z	Z	3/3	−0.100	−
Speed change Z	Z	3/3	−0.124	−
Speed-slope Z	Z	3/3	−0.075	−
Previous Q3- prior mean	History	3/3	0.053	+

Previous Q3 showed the largest positive coefficient,

indicating that recent stress history was the strongest stable predictor. Previous Q1 and sleep-history variables were also repeatedly selected. Light and personalized movement features provided additional contextual information. These coefficients represent predictive associations and should not be interpreted causally.

F. Cross-User Generalization

Nested leave-one-user-out validation was conducted using 400 observations from participants with at least five prior days of history.

TABLE VIII. NESTED LEAVE-ONE-USER-OUT VALIDATION RESULTS

Model	Pooled AUC	Mean user AUC	Bal. Acc.	Specificity	Log Loss
Raw	0.381	0.494	0.436	0.149	0.676
Raw+Z+History	0.643	0.619	0.625	0.571	0.643

The below-chance pooled ROC-AUC of the Raw model should be interpreted together with its mean user-level ROC-AUC of 0.494. Participant-level performance was generally close to chance, whereas differences in probability scales across the separately fitted LOSO models produced poorer ranking when predictions from all participants were pooled. This pattern also suggests substantial between-participant heterogeneity in absolute sensor features rather than a consistently reversed prediction direction for every participant.

The proposed model improved pooled ROC-AUC from 0.381 to 0.643 and mean user-level ROC-AUC from 0.494 to 0.619. Participant-clustered bootstrap analysis showed an ROC-AUC difference of 0.262 with a 95% confidence interval of 0.138~0.357, and ROC-AUC improved for 8 of the 10 held-out participants.

Log Loss decreased from 0.676 to 0.643, but the corresponding confidence interval included zero. Platt scaling also did not improve LOSO Log Loss. Therefore, the proposed model improved cross-user discrimination, while probability reliability for held-out users remained limited. Because personal history was available, this evaluation does not represent a fully history-free cold-start setting.

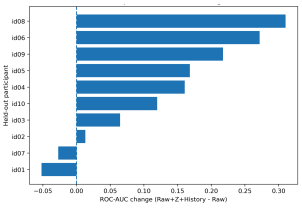


Fig. 5. Participant-level ROC-AUC (left)

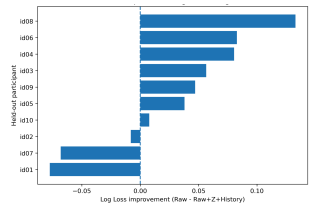


Fig. 6. Participant-level Log Loss (right)

VI. Conclusion

The proposed model improved both chronological and cross-user stress discrimination by combining raw pre-sleep behavior, personalized deviations, and recent questionnaire and sleep history. History features explained most of the performance gain, while Z-score features provided complementary information.

In practice, the model could be integrated into a smartphone-based sleep or wellness service. For example, when pre-sleep screen use, movement, or ambient-light patterns deviate sharply from a user's usual baseline, the system could recommend reducing screen exposure or provide a brief relaxation or breathing prompt.

The main limitations are the small cohort of only 10 participants, incomplete sensor availability, dependence on accumulated personal history, and the relatively high feature-to-sample ratio. Although 400 observations were available, the final candidate pool contained 93 features, increasing the risk of overfitting and unstable feature selection. Training-fold feature reduction, regularization, and nested validation were used to mitigate these risks; however, participant-level estimates and clustered-bootstrap confidence intervals remained sensitive to individual users. Therefore, the reproducibility and generalizability of the LOSO results remain limited. Future work should validate the model in larger and more diverse populations and evaluate whether personalized interventions improve sleep and well-being. Future work should also investigate transformer-based and multimodal deep learning models while maintaining interpretability through explainable AI techniques.

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